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Component and communication system

Abstract

A component is configured to be mounted on a human-powered vehicle. The component includes a communication device. The communication device is configured to receive predetermined information sent from a transmission device of the human-powered vehicle. The predetermined information includes at least some of first information that the transmission device receives from an operating device mounted to the human-powered vehicle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application U.S. application Ser. No. 17/360,175, filed Jun. 28, 2021. The entire disclosure of U.S.

application Ser. No. 17/360,175 is hereby incorporated herein by reference. This application claims priority to Japanese Patent Application No. 2020-116591, filed on Jul. 6, 2020. The entire disclosure of Japanese Patent Application No. 2020-116591 is hereby incorporated herein by reference.

BACKGROUND

Technical Field

(1) The present invention generally relates to a component and a communication system for human-powered vehicles.

Background Information

(2) Some human-powered vehicles are provided with a communication device that is configured to perform communications between components. Japanese Laid-Open Patent Publication No. 2018-39466 discloses an example of a bicycle component including a communication device configured to perform communication in response to operation of an operating portion.

SUMMARY

(3) One object of the present disclosure is to provide a component and a communication system that perform appropriate communication in human-powered vehicles.

(4) In accordance with a first aspect of the present disclosure, a component that is configured to be mounted on a human-powered vehicle. The component includes a communication device. The communication device is configured to receive predetermined information sent from a transmission device of the human-powered vehicle. The predetermined information includes at least some of first information that the transmission device receives from an operating device mounted to the human-powered vehicle. The component according to the first aspect allows at least some of information that the transmission device receives from an operating device mounted to the human-powered vehicle to be transmitted to the communication device.

(5) In accordance with a second aspect of the present disclosure, the component according to the first aspect is configured so that the communication device is configured to receive the predetermined information on a wireless signal. The component according to the second aspect allows the communication device to wirelessly receive predetermined information.

(6) In accordance with a third aspect of the present disclosure, the component system according to the first or second aspect further comprises a motor unit configured to apply a propulsion force to the human-powered vehicle. The component according to the third aspect can apply a propulsion force to the human-powered vehicle.

(7) In accordance with a fourth aspect of the present disclosure, the component according to any one of the first aspect to the third aspect is configured so that the operating device is configured to communicate with the transmission device using a wireless signal. The component according to the fourth aspect performs appropriate wirelessly communication with the transmission device.

(8) In accordance with a fifth aspect of the present disclosure, the component according to any one of the first aspect to the fourth third aspect is configured so that the communication device is electrically connected to the transmission device by a wire to communicate with the transmission device using a wire signal. The component according to the fifth aspect performs appropriate wired communication with the transmission device.

(9) In accordance with a sixth aspect of the present disclosure, the component according to any one of the first aspect to the fifth aspect is configured so that the first information includes at least one of information related to the transmission device and information related to the component. The component according to the sixth aspect performs appropriate communication.

(10) In accordance with a seventh aspect of the present disclosure, the component according to any one of the first aspect to the sixth aspect is configured so that the predetermined information includes at least one of information related to the transmission device and information related to the component. In the component according to the seventh aspect performs appropriate

communication.
(11) In accordance with an eighth aspect of the present disclosure, the component according to any one of the first aspect to the seventh aspect is configured so that the first information conforms to the predetermined information. In the component according to the eighth aspect performs appropriate communication.
(12) In accordance with a ninth aspect of the present disclosure, the component according to any one of the first aspect to the eighth aspect is configured so that the component is configured to receive a signal sent from the operating device. In the component according to the ninth aspect, the component can receive a signal sent from the operating device.
(13) In accordance with a tenth aspect of the present disclosure, a communication system comprises the component according to any one of the first aspect to the ninth aspect and further comprises at least one battery configured to supply electric power to at least one of the component, the operating device and the transmission device. In the communication system according to the tenth aspect at least one of the component, the operating device and the transmission device can receive electric power from at least one battery.
(14) In accordance with an eleventh aspect of the present disclosure, the communication system according to the tenth aspect is configured the battery is configured to supply electric power to the component and the transmission device. In the communication system according to the eleventh aspect, at least the component and the transmission device can receive electric power from the battery.
(15) In accordance with a twelfth aspect of the present disclosure, the communication system according to the eleventh aspect is configured so that the battery is configured to supply electric power to the component and the operating device. In the communication system according to the twelfth aspect, at least the component and the transmission device can receive electric power from the battery.
(16) In accordance with a thirteenth aspect of the present disclosure, the communication system according to any one of the ten aspect to the twelfth aspect is configured so that the battery includes a first configured to supply electric power to the transmission device and the component and a third battery configured to supply electric power to the operating device. In the communication system according to the thirteenth aspect, the electronic controller appropriately controls at least one of the operating device, electric power can be appropriately supplied to the transmission device and the component a first battery and power can be appropriately supplied to a the operating device using a third battery.
(17) The component and the communication system according to the present disclosure perform appropriate communication.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Referring now to the attached drawings which form a part of this original disclosure.
(2) FIG. **1** is a side elevational view of a human-powered vehicle including a communication system, a component and a control system in accordance with a first embodiment.
(3) FIG. **2** is a block diagram showing an electrical connection relationship between the control device and various components in accordance with the first embodiment.
(4) FIG. **3** is a flowchart showing an example of control executed by an electronic controller of the first embodiment.
(5) FIG. **4** is a block diagram showing an electrical connection relationship between a control device and various components in accordance with a second embodiment.
(6) FIG. **5** is a flowchart showing an example of control executed by a first electronic controller of

the second embodiment.
(7) FIG. **6** is a flowchart showing an example of control executed by a second electronic controller of the second embodiment.
(8) FIG. **7** is a flowchart showing an example of control executed by a third electronic controller of the second embodiment.
(9) FIG. **8** is a block diagram showing an electrical connection relationship between a control device and various components in accordance with a third embodiment.
(10) FIG. **9** is a block diagram showing an electrical connection relationship between a control device and various components in accordance with a fourth embodiment.
(11) FIG. **10** is a block diagram showing an electrical connection relationship between a control device and various elements in a modified embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

(12) Selected embodiments will now be explained with reference to the drawings. It will be apparent to those skilled in the bicycle field from this disclosure that the following descriptions of the embodiments are provided for illustration only and not for the purpose of limiting the invention as defined by the appended claims and their equivalents.

First Embodiment

(13) A human-powered vehicle A including a communication system **10** will now be described with reference to FIGS. **1** to **3**. A human-powered vehicle refers to a vehicle that at least partially uses human force as a prime mover to travel, and includes vehicles that assist human force with electric power. The human-powered vehicle does not include vehicles using only a prime mover that is not human force. In particular, the human-powered vehicle does not include vehicles that use only an internal combustion engine as the prime mover. The human-powered vehicle is generally assumed to be a small, light vehicle that does not typically require a license for driving on a public road. More specifically, the illustrated human-powered vehicle A is a trekking bike. The human-powered vehicle A further includes a frame A**1**, a front fork A**2**, a wheel W, a handlebar H, and a drivetrain B. The wheel W includes a front wheel WF and a rear wheel WR.
(14) The drivetrain B is of, for example, a chain-drive type. The drivetrain B includes a crank C, a front sprocket assembly D**1**, a rear sprocket assembly D**2**, and a chain D**3**. The crank C includes a crank axle C**1** rotatably supported by the frame A**1** and two crank arms C**2** respectively provided on opposite ends of a crank axle C**1**. A pedal PD is rotatably coupled to the distal end of each crank arm C**2**. The drivetrain B can be of any type such as a belt-drive type or a shaft-drive type.
(15) The front sprocket assembly D**1** is provided on the crank C to rotate integrally with the crank axle C**1**. The rear sprocket assembly D**2** is provided on a hub HR of the rear wheel WR. The chain D**3** runs around the front sprocket assembly D**1** and the rear sprocket assembly D**2**. A human driving force applied to the pedals PD by an occupant who is riding the human-powered vehicle A is transmitted via the front sprocket assembly D**1**, the chain D**3**, and the rear sprocket assembly D**2** to the rear wheel WR.
(16) The communication system **10** includes a first communication device **12**, a second communication device **14** and a third communication device **16**. Unless otherwise expressly limited, the term “communication device” as used herein refers to hardware capable of transmitting and/or receiving an analog or digital signal either by a wire or wirelessly. The term “communication device” does not include a human. The term “wireless communication device” as used herein refers to hardware capable of transmitting and/or receiving an analog or digital signal wirelessly, while the term “wired communication device” as used herein refers to hardware capable of transmitting and/or receiving an analog or digital signal via a wire.
(17) The first communication device **12** is configured to be connected to an operating device **20** mounted on the human-powered vehicle A. The second communication device **14** is configured to be connected to a transmission device **30** of the human-powered vehicle A. The third communication device **16** is configured to be connected to a component **40** of the human-powered

vehicle A differing from the transmission device **30**. The third communication device **16** is connected to the component **40** that configured to be mounted on the human-powered vehicle A and that differs from the operating device **20** and the transmission device **30**. The operating device **20**, the transmission device **30**, and the component **40** are mounted on the single human-powered vehicle A. The component **40** has a configuration mounted on the human-powered vehicle A that differs from the operating device **20** and the transmission device **30**. In the communication system **10**, the second communication device **14** receives first information that is sent from the first communication device **12** in response to operation of the operating device **20**, and the third communication device **16** of the component **40** receives second information that includes at least some of the first information sent from the second communication device **14**. Thus, in this way, the third communication unit device **16** is configured to receive predetermined information sent from the transmission device **30** of the human-powered vehicle A.

(18) The first communication device **12** includes at least a transmitter configured to transmit a signal that includes information. Preferably, the first communication device **12** also includes a receiver configured to receive a signal that includes information. The second communication device **14** includes both a transmitter configured to transmit a signal that includes information and a receiver configured to receive a signal that includes information. The third communication device **16** includes at least a receiver configured to receive a signal that includes information. Preferably, the third communication device **16** also includes a transmitter configured to transmit a signal that includes information. The first communication device **12** and the second communication device **14** are connected to perform wired or wireless communication with each other. The first communication device **12** and the second communication device **14** communicate with each other through, for example, power line communication (PLC), a controller area network (CAN), or a universal asynchronous receiver/transmitter (UART). The second communication device **14** and the third communication device **16** are connected to perform wired or wireless communication with each other.

(19) The communication system **10** further includes a battery BT that is configured to supply electric power to at least one of the operating device **20**, the transmission device **30**, and the component **40**. The operating device **20** and the component **40** are connected by a power line. The operating device **20** and the battery BT can be connected by a power line. The battery BT includes one or more battery elements. The battery elements include a rechargeable battery.

(20) The human-powered vehicle A also includes a control device **50**. The control device **50** includes an electronic controller **52** and storage **54**. As explained below, the electronic controller **52** can control the communication system **10**, the transmission device **30** and the component **40**. The storage **54** stores various types of information. The control device **50** and the communication system **10** constitutes a control system CS for the human-powered vehicle A. The electronic controller **52** is formed of one or more semiconductor chips that are mounted on a circuit board. The term “electronic controller” as used herein refers to hardware that executes a software program, and does not include a human. Hereinafter, the electronic controller **52** will be simply referred to as the controller **52**.

(21) The battery BT is connected to the controller **52** of the control device **50** to perform wired or wireless communication with the controller **52**. The battery BT and the controller **52** of the control device **50** can be connected by connection terminals. In a case where the controller **52** of the control device **50** is provided on the battery BT, the third communication device **16** of the component **40** and the controller **52** can be wire-connected. The battery BT can include a first battery BT**1** that is configured to supply electric power to the operating device **20** and the component **40**, and the second battery BT**2** that is configured to supply electric power to the transmission device **30**. At least one of the first battery BT**1** and the second battery BT**2** can be configured to supply electric power to all of the operating device **20**, the transmission device **30**, and the component **40**. Thus, the first battery BT**1** is configured to supply electric power to the

component **40** and the operating device **20**. The operating device **20** and the transmission device **30** can be connected by a power line. In a case where the operating device **20** is not connected to the first battery BT**1** and the second battery BT**2** by a power line, the battery BT can further include a third battery BT**3** that is configured to supply electric power to the operating device **20**. The third battery BT**3** supplies electric power to the operating device **20**, for example, in a case where the operating device **20** performs wireless communication with the transmission device **30**.

(22) In an example, the first battery BT**1** and the second battery BT**2** have different power capacities. In a case where the transmission device **30** is electrically connected to the first battery BT**1** and the second battery BT**2**, the controller **52** can be configured to give priority to the battery BT having the larger power capacity for supplying electric power to the transmission device **30**. In a case where the component **40** is electrically connected to the first battery BT**1** and the second battery BT**2**, the controller **52** can be configured to give priority to the battery BT having the larger power capacity for supplying electric power to the component **40**. In an example, the power capacity of the first battery BT**1** is larger than the power capacity of the second battery BT**2**. The transmission device **30** and the component **40** can be electrically connected to the first battery BT**1**. In this configuration, in a case where the amount of power in the first battery BT**1** is less than a predetermined value, the priority for being supplied with electric power can be given to one of the transmission device **30** and the component **40**. Preferably, the first battery BT**1** and the second battery BT**2** are connected by a power line. The first battery BT**1** and the second battery BT**2** are configured so that in a case where the amount of power in one of the first battery BT**1** and the second battery BT**2** is less than the predetermined value, the power is supplied from the other one of the first battery BT**1** and the second battery BT**2**.

(23) The human-powered vehicle A includes the operating device **20**, the transmission device **30**, and the component **40**, which differs from the operating device **20** and the transmission device **30**. The component **40** further comprises a motor unit **42** configured to apply a propulsion force to the human-powered vehicle A. Preferably, the component **40** further includes at least one of a brake device **44**, a suspension **46**, and an adjustable seatpost **48**.

(24) The operating device **20** outputs a predetermined signal in accordance with operation of the user. The operating device **20** is provided on a steering mechanism of the human-powered vehicle A. The steering mechanism includes the handlebar H. The operating device **20** includes at least a shift operating device that operates the transmission device **30**. The shift operating device includes at least one of a shifter and a switch. The shift operating device outputs a signal including information related to shifting that controls the transmission device **30** in accordance with operation of the user. The shift operating device sends a signal including information related to shifting via the first communication device **12**. In other words, as seen in FIG. **2**, the operating device **20** is configured to communicate with the transmission device **30** using a wireless signal. The operating device **20** further includes a component operating device that operates the component **40** of the human-powered vehicle A. In a case where the component **40** includes the motor unit **42**, the component operating device sends a signal including information related to an output of a motor in relation to human driving force via the first communication device **12**. In a case where the component **40** includes the brake device **44**, the component operating device sends a signal including information related to braking via the first communication device **12**. In a case where the component **40** includes the suspension **46**, the component operating device sends a signal including information related to actuation of the suspension **46** via the first communication device **12**. In a case where the component **40** includes the adjustable seatpost **48**, the component operating device sends a signal including information related to height of the adjustable seatpost **48** via the first communication device **12**. The first communication device **12** connected to the operating device **20** can be a single first communication device **12** connected to two or more operating devices **20** or can be two or more first communication units **12** connected to two or more operating devices **20**.

(25) The transmission device **30** includes an external transmission device. The external transmission device is configured to change a transmission ratio, which is defined by at least one of the number of teeth and the diameter of each of a front sprocket and a rear sprocket around which the chain D**3** runs. The front sprocket is part of the front sprocket assembly D**1** including one or more front sprockets. The rear sprocket is part of the rear sprocket assembly D**2** including one or more rear sprockets. In an example, the transmission device **30** includes at least one of a front derailleur **32** and a rear derailleur **34**. The front derailleur **32** is provided in the vicinity of the front sprocket assembly D**1**. The front derailleur **32** is actuated to change the front sprocket on which the chain D**3** runs, thereby changing transmission ratio. The transmission ratio is determined based on the relationship between the number of teeth on the front sprocket and the number of teeth on the rear sprocket. In an example, the transmission ratio is determined by the ratio of a rotational speed of the rear sprocket to a rotational speed of the front sprocket. More specifically, the transmission ratio is defined by the ratio of the number of teeth on the front sprocket to the number of teeth on the rear sprocket. The rear derailleur **34** is provided on a rear end A**3** of the frame A**1**. The rear derailleur **34** is actuated to change the rear sprocket on which the chain D**3** runs, thereby changing the transmission ratio. The transmission device **30** can include an internal transmission device instead of the external transmission device. In this case, the internal transmission device is provided on, for example, the hub HR of the rear wheel WR. The transmission device **30** can include a stepless transmission device instead of the external transmission device. In this case, the stepless transmission device is provided on, for example, the hub HR of the rear wheel WR. The transmission device **30** changes the transmission ratio of the human-powered vehicle A in accordance with an operating signal from the shift operating device.
(26) The motor unit **42** is actuated to assist in propulsion of the human-powered vehicle A. The motor unit **42** is actuated in accordance with, for example, human driving force applied to the pedals PD. The motor unit **42** includes a motor such as an electric motor. The motor unit **42** is actuated by electric power supplied from the battery BT mounted on the human-powered vehicle A. The component operating device includes a motor unit operating device that operates the motor unit **42**. The motor unit operating device outputs an operating signal that changes a control state of the motor unit **42** and assisting force.
(27) The brake device **44** includes brake devices **44**, the number of which corresponds to the number of wheels W. The brake device **44** includes a brake device **44** corresponding to the front wheel WF and a brake device **44** corresponding to the rear wheel WR. The two brake devices **44** can have the same configuration. Each brake device **44** is, for example, a rim brake device that brakes a rim of the human-powered vehicle A. The component operating device incudes a brake operating device. Each brake device **44** is electrically driven based on a signal that is output in response to corresponding operation of the brake operating device. Each brake device **44** can be a disc brake device that brakes a disc brake rotor mounted on the human-powered vehicle A.
(28) The suspension **46** includes at least one of a front suspension and a rear suspension. The front suspension is actuated to dampen an impact that the front wheel WF receives from the ground. The rear suspension is actuated to dampen an impact that the rear wheel WR receives from the ground. The component operating device includes a suspension operating device. Each suspension **46** is electrically driven in accordance with corresponding operation of the suspension operating device. More specifically, at least one of the movement state, travel amount, damping force, and repulsive force of each suspension **46** is changed based on an operating signal that is output in response to corresponding operation of the suspension operating device.
(29) The adjustable seatpost **48** is actuated to change the height of a saddle relative to the frame A**1**. The component operating device includes an adjustable seatpost operating device. The adjustable seatpost **48** is electrically driven based on an operating signal that is output in response to operation of the adjustable seatpost operating device.
(30) Since the control device **50** is used to control the first communication device **12**, the second

communication device **14**, and the third communication device **16**, the control device **50** can be considered to be part of the communication system **10**. Stated differently, the communication system **10** further includes the control device **50** that includes the controller **52**, which is configured to control the first communication device **12**, the second communication device **14**, and the third communication device **16**. The controller **52** controls at least one of the operating device **20** mounted on the human-powered vehicle A, the transmission device **30**, and the component **40** mounted on the human-powered vehicle A.

(31) The controller **52** includes at least one of a central processing device (CPU) and a micro processing device (MPU), Thus, the controller **52** includes at least one processor and at least one computer storage device (i.e., computer memory devices). The controller **52** can also control the transmission device **30** in accordance with operation of the shift operating device. The control device **50** further includes the storage **54** that stores various types of information. The storage **54** is any computer storage device or any non-transitory computer-readable medium with the sole exception of a transitory, propagating signal. For example, the storage **54** includes a nonvolatile memory and a volatile memory. The nonvolatile memory includes, for example, at least one of a read-only memory (ROM), an erasable programmable read only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), and a flash memory. The volatile memory includes, for example, a random access memory (RAM). The storage **54** stores, for example, various types of programs used for control and predetermined information.

(32) In a first example, the control device **50** is provided on at least the transmission device **30**. The control device **50** is provided on at least one of the front derailleur **32** and the rear derailleur **34**. The controller **52** receives the first information via the second communication device **14** through wired or wireless communication. The controller **52** is configured so that the second communication device **14** sends a signal including the second information through wired or wireless communication and so that the third communication device **16**, which is provided on the component **40**, receives the signal sent from the second communication device **14**. In the first example, each of the operating device **20** and the component **40** includes an electronic controller that differs from the controller **52** of the control device **50**. In a second example, the control device **50** is provided on the component **40**. In a case where the component **40** includes the motor unit **42**, the control device **50** is provided, for example, on the motor unit **42**. In an example, the control device **50** is accommodated in a housing **42**A of the motor unit **42**. In the second example, each of the operating device **20** and the transmission device **30** includes an electronic controller that differs from the controller **52** of the control device **50**. The controller **52** receives a signal including the second information via the third communication device **16** through wired or wireless communication. In a third example, the control device **50** is provided on the battery BT. In the third example, the controller **52** can be arranged in a housing of the battery BT and be connected to the battery BT by terminals. In the third example, the controller **52** controls the battery BT and the component **40** based on information received by the third communication device **16**. The controller **52** can be configured to send information related to control to an electronic controller that differs from the controller **52** provided on the component **40** through wired or wireless communication, and the controller **52** can be configured to process information included in a signal received by the third communication device **16** provided on the component **40** and send the signal to a further controller of the component **40**. In this case, a signal that includes third information can be sent to the third communication device **16** provided on the component **40** based on a signal from the controller **52**. The control device **50** is actuated by electric power supplied from the battery BT. The control device **50** is configured to perform wired or wireless communication with at least one of the first communication device **12**, the second communication device **14**, and the third communication device **16**. FIG. **2** discloses the configuration of the first example of the control device **50**.

(33) The controller **52** controls the first communication device **12** to output the first information in response to operation of the operating device **20**. The controller **52** controls the second

communication device **14** to receive the first information and output the second information including at least some of the first information. The controller **52** controls the third communication device **16** to receive the second information.

(34) The first information and the second information will be described. The first information and the second information include at least one of information related to the transmission device **30** and information related to the component **40**. The information related to the transmission device **30** includes at least one of information related to a control state of the transmission device **30** and information for the controller **52** to control the transmission device **30**. The information related to the control state of the transmission device **30** includes information of the present transmission ratio. The information for the controller **52** to control the transmission device **30** includes information of an operating signal related to the changing of the transmission ratio. The information related to the component **40** includes at least one of information related to a control state of the component **40** and information for the controller **52** to control the component **40**. In a case where the component **40** includes the motor unit **42**, information related to the motor unit **42** includes at least one of information related to a control state of the motor unit **42** and information for the controller **52** to control the motor unit **42**. The control state of the motor unit **42** includes a first control state and a second control state that differs in assisting force from the first control state. Assisting force includes at least one of a maximum output value of the motor, a minimum output value of the motor, a maximum output torque of the motor, a minimum output torque of the motor, a response speed of the motor, a ratio of a motor output to human torque that is input to the human-powered vehicle A, and driving force of the motor in relation to human driving force that is input to the human-powered vehicle A. The information for the controller **52** to control the motor unit **42** includes information related to assisting force.

(35) In a first example, the first information conforms to the second information. In an example, the first information and the second information include the information related to the transmission device **30** and the information related to the component **40**. In another example, the first information and the second information include one of the information related to the transmission device **30** and the information related to the component **40**. In a second example, the first information differs from the second information. The controller **52** is configured to change the contents of the second information. In an example, the first information includes the information related to the transmission device **30** and the information related to the component **40**, and the second information includes only the information related to the component **40**. In another example, the first information includes only the information related to the transmission device **30**, and the second information includes the information related to the transmission device **30** and the information related to the component **40**.

(36) A sending process of the first information and the second information executed by the controller **52** will be described. The controller **52** sends the first information to the second communication device **14** on a wireless signal or a wire signal. The controller **52** sends the first information from the transmitter of the first communication device **12**. The controller **52** sends the first information to the second communication device **14** on a wire signal in a case where the operating device **20** and the transmission device **30** are wire-connected. The controller **52** outputs the first information on a wireless signal in a case where the operating device **20** and the transmission device **30** are not wire-connected. In other words, in this way, the operating device **20** is configured to communicate with the transmission device **30** using a wireless signal.

(37) The controller **52** sends the second information to the third communication device **16** on a wireless signal or a wire signal. The controller **52** sends the second information from the transmitter of the second communication device **14**. The controller **52** sends the second information to the third communication device **16** on a wire signal in a case where the transmission device **30** and the component **40** are wire-connected. The controller **52** outputs the second information on a wireless signal in a case where the transmission device **30** and the component **40** are not wire-

connected.

(38) The controller **52** can send any combination of a wire signal and a wireless signal. In a first example, the controller **52** controls the second communication device **14** so that the second communication device **14** receives the first information output from the first communication device **12** on a wire signal and sends the second information output from the second communication device **14** to the third communication device **16** on a wire signal. In a second example, the controller **52** controls the second communication device **14** so that the second communication device **14** receives the first information output from the first communication device **12** on a wireless signal and sends the second information output from the second communication device **14** to the third communication device **16** on a wireless signal. In a third example, the controller **52** controls the second communication device **14** so that the second communication device **14** receives the first information output from the first communication device **12** on a wire signal and sends the second information output from the second communication device **14** to the third communication device **16** on a wireless signal. In a fourth example, the controller **52** controls the second communication device **14** so that the second communication device **14** receives the first information output from the first communication device **12** on a wireless signal and sends the second information output from the second communication device **14** to the third communication device **16** on a wire signal.

(39) Control of the operating device **20**, the transmission device **30**, and the component **40** executed by the controller **52** will now be described. The controller **52** controls the supply of electric power from the battery BT. The controller **52** controls the supply of electric power from the battery BT to at least one of the operating device **20**, the transmission device **30**, and the component **40** based on at least one of the first information and the second information. In an example, in a case where at least one of the first information and the second information includes the information related to the transmission device **30**, the controller **52** supplies electric power to the operating device **20** and the transmission device **30** from the battery BT. In a case where each of the first information and the second information includes the information related to the transmission device **30**, the controller **52** changes the amount of power supplied from the battery BT to at least one of the operating device **20** and the transmission device **30** from that in a case where the information related to the transmission device **30** is not included. In an example, in a case where the first communication device **12** of the operating device **20** and the second communication device **14** of the transmission device **30** are wire-connected and the information related to the transmission device **30** is included, the controller **52** increases the amount of power supplied to the operating device **20** and the transmission device **30** as compared to in a case where the information related to the transmission device **30** is not included. In a case where the first communication device **12** of the operating device **20** and the second communication device **14** of the transmission device **30** are wireless-connected and the information related to the transmission device **30** is included, the controller **52** increases the amount of power supplied to the transmission device **30** as compared to in a case where the information related to the transmission device **30** is not included. While electric power is supplied from the battery BT, the controller **52** controls the electric power to be supplied to the operating device **20**. In a case where the second information includes the information related to the component **40**, the controller **52** supplies electric power to the component **40** from the battery BT. In a case where the second information does not include the information related to the component **40**, the controller **52** can change the amount of power supplied to the component **40** from the battery BT from that in a case where the second information includes the information related to the component **40**.

(40) The controller **52** controls the supply of electric power from the first battery BT**1** and the second battery BT**2** based on at least one of the first information and the second information. In an example, in a case where at least one of the first information and the second information includes the information related to the transmission device **30**, the controller **52** supplies electric power from

the second battery BT**2** to the transmission device **30**. In a case where each of the first information and the second information includes the information related to the transmission device **30**, the controller **52** changes the amount of power supplied to the transmission device **30** from the second battery BT**2** from that in a case where the information related to the transmission device **30** is not included. In an example, in a case where the information related to the transmission device **30** is included, the controller **52** increases the amount of power supplied to the transmission device **30** as compared to in a case where the information related to the transmission device **30** is not included. In a case where the second information includes the information related to the component **40**, the controller **52** supplies electric power from the first battery BT**1** to the component **40**. In a case where the second information does not include the information related to the component **40**, the controller **52** can change the amount of power supplied from the first battery BT**1** to the component **40** from that in a case where the second information includes the information related to the component **40**.
(41) The controller **52** controls the transmission device **30** and the component **40** based on at least one of the first information and the second information. In an example, in a case where at least one of the first information and the second information includes information for controlling the transmission device **30**, the transmission device **30** is controlled based on the information for controlling the transmission device **30**. In a case where the second information includes information for controlling the component **40**, the controller **52** controls the component **40** based on the information for controlling the component **40**.
(42) An example of control executed by the controller **52** of the first embodiment will now be described with reference to FIG. **3**. While electric power is supplied from the battery BT, the controller **52** cyclically executes the process from steps S**11** to S**18**.
(43) In step S**11**, the controller **52** determines whether the operating device **20** is operated. In a case where the determination is affirmative, the controller **52** executes step S**12**. In a case where the determination is negative, the controller **52** again executes step S**11**.
(44) In step S**12**, the controller **52** determines whether the operating device **20** and the transmission device **30** are wire-connected. The controller **52** can, for example, detect which one of the batteries BT is supplying electric power to the operating device **20** to determine whether the operating device **20** and the transmission device **30** are wire-connected. In a case where the determination is affirmative, the controller **52** executes step S**13**. In a case where the determination is negative, the controller **52** executes step S**14**.
(45) In step S**13**, the controller **52** outputs the first information from the first communication device **12** on a wire signal. Subsequent to step S**13**, the controller **52** executes step S**15**. In step S**14**, the controller **52** outputs the first information from the first communication device **12** on a wireless signal. Subsequent to step S**14**, the controller **52** executes step S**15**.
(46) In step S**15**, the controller **52** determines whether the transmission device **30** and the component **40** are wire-connected. In a case where the determination is affirmative, the controller **52** executes step S**16**. In a case where the determination is negative, the controller **52** executes step S**17**.
(47) In step S**16**, the controller **52** outputs the second information from the second communication device **14** on a wire signal. Subsequent to step S**16**, the controller **52** executes step S**18**. In step S**17**, the controller **52** outputs the second information from the second communication device **14** on a wireless signal. Subsequent to step S**17**, the controller **52** executes step S**18**.
(48) In step S**18**, the controller **52** controls the transmission device **30** and the component **40** based on at least one of the first information and the second information. Subsequent to step S**18**, the controller **52** ends the control.
Second Embodiment
(49) A second embodiment of the communication system **10** and the control device **50** will be described with reference to FIGS. **4** to **7**. The communication system **10** and the control device **50**

of the second embodiment is the same as the communication system **10** and the control device **50** of the first embodiment except for the configurations of the controller **52** and the battery BT. Same reference characters are given to those configurations that are the same as the corresponding configurations of the first embodiment. Such configurations will not be described in detail.
(50) As shown in FIG. **4**, the controller **52** includes a first controller **52**A, a second controller **52**B, and a third controller **52**C. The first controller **52**A controls the operating device **20**, the first communication device **12** connected to the operating device **20**, and a notification device **56**. The second controller **52**B controls the transmission device **30** and the second communication device **14** connected to the transmission device **30**. The third controller **52**C controls the component **40** and the third communication device **16** connected to the component **40**.
(51) The first controller **52**A is provided on the operating device **20**. The first controller **52**A outputs the first information in response to operation of the operating device **20**. In a case where the first communication device **12** receives the third information, the first controller **52**A controls the notification device **56** so that the third information is indicated on the notification device **56**.
(52) The notification device **56** includes an indicator. The indicator includes, for example, an indication panel. The notification device **56** includes, for example, at least one of a portable electronic device, a display, a smartphone, a tablet computer, and a cycle computer. The notification device **56** can include a speaker. The notification device **56** is configured to perform wired or wireless communication with at least the first communication device **12**.
(53) The battery BT further includes the third battery BT**3** that supplies electric power to at least one of the first communication device **12**, the operating device **20**, the first controller **52**A, and the notification device **56**. The third battery BT**3** has a smaller power capacity than the first battery BT**1** and the second battery BT**2**. In an example, the third battery BT**3** includes a lithium-ion battery. In another example, the third battery BT**3** includes a button cell or coin battery.
(54) The second controller **52**B is provided on the transmission device **30**. In an example, the second controller **52**B is provided on one of the front derailleur **32** and the rear derailleur **34**. The second controller **52**B controls the transmission device **30**. In a case where the first information is received, the second controller **52**B outputs the second information to the third communication device **16**. In a case where the third information is received, the second controller **52**B outputs the third information to the first communication device **12**.
(55) The third controller **52**C is provided on the component **40**. In a case where multiple components **40** are provided on the human-powered vehicle A, the third controller **52**C can be provided on each component **40** or one of the components **40**. The third controller **52**C controls the component **40**. The third controller **52**C sends the third information from the third communication device **16**. In a case where the third communication device **16** performs wired communication with the second communication device **14**, the third controller **52**C outputs the third information from the third communication device **16** to the second communication device **14**. The third information includes information related to a state of the component **40**. In an example, in a case where the component **40** includes the motor unit **42**, information related to the control state of the motor and assisting force is included. In a case where the component **40** includes the brake device **44**, information related to braking force is included. In a case where the component **40** includes the suspension **46**, information related to the movement state, travel amount, damping force, and repulsive force is included. In a case where the component **40** includes the adjustable seatpost **48**, information related to the height of the saddle relative to the frame A**1** is included.
(56) An example of control executed by the first controller **52**A will now be described with reference to FIG. **5**. While electric power is supplied from the battery BT, the first controller **52**A repeatedly executes the process from steps S**21** to S**26**.
(57) In step S**21**, the first controller **52**A determines whether the operating device **20** is operated. In a case where the determination is affirmative, the first controller **52**A executes step S**22**. In a case where the determination is negative, the first controller **52**A executes step S**25**.

(58) The process from steps S**22** to S**24** executed by the first controller **52**A is the same as the process from steps S**12** to S**14** executed by the controller **52** of the first embodiment.
(59) In step S**25**, the first controller **52**A determines whether the third information is received. In a case where the determination is affirmative, the first controller **52**A executes step S**26**. In a case where the determination is negative, the first controller **52**A ends the control.
(60) In step S**26**, the first controller **52**A outputs the third information to the notification device **56**. Subsequent to step S**26**, the first controller **52**A ends the control.
(61) An example of control executed by the second controller **52**B will now be described with reference to FIG. **6**. While electric power is supplied from the battery BT, the second controller **52**B repeatedly executes the process from steps S**31** to S**37**.
(62) In step S**31**, the second controller **52**B determines whether the first information is received. In a case where the determination is affirmative, the second controller **52**B executes step S**32**. In a case where the determination is negative, the second controller **52**B executes step S**36**.
(63) The process from steps S**32** to S**34** executed by the second controller **52**B is the same as the process from steps S**15** to S**17** executed by the controller **52** of the first embodiment.
(64) In step S**35**, the second controller **52**B controls the transmission device **30** based on the first information. Subsequent to step S**35**, the second controller **52**B executes step S**36**.
(65) In step S**36**, the second controller **52**B determines whether the third information is received. In a case where the determination is affirmative, the second controller **52**B executes step S**37**. In a case where the determination is negative, the second controller **52**B ends the control.
(66) In step S**37**, the second controller **52**B outputs the third information from the second communication device **14** to the first communication device **12**. Subsequent to step S**37**, the second controller **52**B ends the control.
(67) An example of control executed by the third controller **52**C will now be described with reference to FIG. **7**. While electric power is supplied from the battery BT, the third controller **52**C repeatedly executes the process from steps S**41** to S**44**.
(68) In step S**41**, the third controller **52**C determines whether the second information is received. In a case where the determination is affirmative, the third controller **52**C executes step S**42**. In a case where the determination is negative, the third controller **52**C ends the control.
(69) In step S**42**, the third controller **52**C determines whether the information related to the component **40** is included. In a case where the determination is affirmative, the third controller **52**C executes step S**43**. In a case where the determination is negative, the third controller **52**C ends the control.
(70) In step S**43**, the third controller **52**C controls the component **40** based on the second information. Subsequent to step S**43**, the third controller **52**C executes step S**44**.
(71) In step S**44**, the third controller **52**C outputs the third information from the third communication device **16** toward the second communication device **14**. Subsequent to step S**44**, the third controller **52**C ends the control.
Third Embodiment
(72) A third embodiment of the communication system **10** and the control device **50** will be described with reference to FIG. **8**. The communication system **10** and the control device **50** of the third embodiment further includes a fourth communication device **18** provided on the transmission device **30**. The same reference characters are given to those configurations that are the same as the corresponding configurations of the first and second embodiments. Such configurations will not be described in detail.
(73) The communication system **10** further includes the fourth communication device **18**. The fourth communication device **18** is provided on the transmission device **30**. The fourth communication device **18** includes at least a receiver configured to receive a signal that includes information. Preferably, the fourth communication device **18** includes a transmitter configured to transmit a signal that includes information. The fourth communication device **18** is provided on one

of the front derailleur **32** and the rear derailleur **34** of the transmission device **30**, and the second communication device **14** is provided on the other one of the front derailleur **32** and the rear derailleur **34**. In an example, the second communication device **14** is provided on the rear derailleur **34**, and the fourth communication device **18** is provided on the front derailleur **32**. The fourth communication device **18** is configured to perform wired or wireless communication with the third communication device **16**. The fourth communication device **18** can be configured to perform wired or wireless communication with at least one of the first communication device **12** and the second communication device **14**. In an example, the fourth communication device **18** is configured to communicate with the second communication device **14**. In a case where the second communication device **14** is configured to communicate with the third communication device **16** and the fourth communication device **18**, the third communication device **16** and the fourth communication device **18** can be configured not to communicate with each other.

(74) Transmission of a signal including information in the communication system **10** of the third embodiment will now be described. Transmission and reception performed by each communication device is controlled by the controller **52** of the control device **50** or a controller (not shown) provided on each of the operating device **20**, the transmission device **30**, and the component **40**.

(75) The first communication device **12** performs wired or wireless communication with the second communication device **14** provided on the rear derailleur **34**. In an example, the first communication device **12** performs wireless communication with the second communication device **14**. The first communication device **12** sends a signal including first information to the second communication device **14**. The first information includes at least one of information related to the transmission device **30** and information related to the component **40**.

(76) The second communication device **14** performs wired or wireless communication with the third communication device **16** provided on the component **40**. In an example, the second communication device **14** performs wired communication with the third communication device **16**. The second communication device **14** sends a signal including second information to the third communication device **16**. The second information includes at least one of information related to the transmission device **30** and information related to the component **40**. The second communication device **14** can send the second information to the fourth communication device **18** through the electrically connected second battery BT**2** and a power line. The second battery BT**2** further includes a communication device that includes both a transmitter configured to transmit a signal including information and a receiver configured to receive a signal including information.

(77) The third communication device **16** communicates with at least one of the fourth communication device **18** provided on the front derailleur **32** and the second communication device **14**. The third communication device **16** performs wired or wireless communication with the fourth communication device **18**. In an example, the third communication device **16** performs wired communication with the fourth communication device **18**. In a case where the received second information includes the information related to the transmission device **30**, the third communication device **16** sends a signal including the information related to the transmission device **30** to at least one of the second communication device **14** and the fourth communication device **18**. In a case where the second communication device **14** communicates with the fourth communication device **18**, the third communication device **16** can be configured not to perform wired and wireless communication with the fourth communication device **18**.

(78) The first battery BT**1** supplies electric power to the component **40** and the third communication device **16**. The second battery BT**2** supplies electric power to at least the front derailleur **32**, the rear derailleur **34**, the second communication device **14**, and the fourth communication device **18**. The third battery BT**3** supplies electric power to the operating device **20** and the first communication device **12**. The third battery BT**3** includes, for example, a button cell or coin battery. In a case where the first communication device **12** and the second communication device **14** are connected by a power line that allows for power line communication, the second

battery BT**2** supplies electric power to the operating device **20** and the first communication device **12**.

Fourth Embodiment

(79) A fourth embodiment of the communication system **10** and the control device **50** will be described with reference to FIG. **9**. The communication system **10** and the control device **50** of the fourth embodiment do not include the second battery BT**2** of the third embodiment. Same reference characters are given to those configurations that are the same as the corresponding configurations of the first to third embodiments. Such configurations will not be described in detail.

(80) Transmission of a signal including information in the communication system **10** of the fourth embodiment will now be described. Transmission and reception performed by each communication device is controlled by the controller **52** of the control device **50** or a controller (not shown) provided on each of the operating device **20**, the transmission device **30**, and the component **40**.

(81) The first communication device **12** performs wired or wireless communication with the second communication device **14** provided on the rear derailleur **34**. In an example, the first communication device **12** performs wireless communication with the second communication device **14**. The first communication device **12** sends a signal including first information to the second communication device **14**. The first information includes at least one of information related to the transmission device **30** and information related to the component **40**.

(82) The second communication device **14** performs wired or wireless communication with the third communication device **16** provided on the component **40**. In an example, the second communication device **14** performs wired communication with the third communication device **16**. The second communication device **14** sends a signal including second information to the third communication device **16**. The second information includes at least one of information related to the transmission device **30** and information related to the component **40**. The second communication device **14** can send the second information to the fourth communication device **18** by a power line that electrically connects the second communication device **14** and the fourth communication device **18**.

(83) The third communication device **16** communicates with at least one of the fourth communication device **18** provided on the front derailleur **32** and the second communication device **14**. The third communication device **16** performs wired or wireless communication with the fourth communication device **18**. In an example, the third communication device **16** performs wired communication with the fourth communication device **18**. In a case where the received second information includes the information related to the transmission device **30**, the third communication device **16** sends a signal including the information related to the transmission device **30** to at least one of the second communication device **14** and the fourth communication device **18**. In a case where the second communication device **14** communicates with the fourth communication device **18**, the third communication device **16** can be configured not to perform wired and wireless communication with the fourth communication device **18**.

(84) The first battery BT**1** supplies electric power to the component **40**, the front derailleur **32**, the rear derailleur **34**, the second communication device **14**, the third communication device **16**, and the fourth communication device **18**. The second battery BT**2** is omitted or integrated with the first battery BT**1**. The third battery BT**3** supplies electric power to the operating device **20** and the first communication device **12**. The third battery BT**3** includes, for example, a button cell or coin battery. In a case where the first communication device **12** and the second communication device **14** are connected by a power line that allows for power line communication, the first battery BT**1** supplies electric power to the operating device **20** and the first communication device **12**.

Modifications

(85) The description related to the above embodiments exemplifies, without any intention to limit, applicable forms of a communication system and a control device according to the present disclosure. The communication system and the control device according to the present disclosure

can be applied to, for example, modifications of the embodiments that are described below and combinations of at least two of the modified examples that do not contradict each other. In the modifications described hereinafter, same reference characters are given to those elements that are the same as the corresponding elements of the above embodiment. Such elements will not be described in detail.

(86) The receiver and the transmitter of the second communication device **14** can be configured separately from each other. In an example, the receiver of the second communication device **14** is provided on the front derailleur **32**, and the transmitter of the second communication device **14** is provided on the rear derailleur **34**. In another example, the receiver of the second communication device **14** is provided on the rear derailleur **34**, and the transmitter of the second communication device **14** is provided on the front derailleur **32**.

(87) The third communication device **16** can be provided on the battery BT. In an example, the third communication device **16** is provided on the first battery BT**1**. The third communication device **16** and the controller **52** configured to control the component **40** are wire-connected.

(88) In the communication system **10** of the third embodiment and the communication system **10** of the fourth embodiment, in a case where the first communication device **12** and the second communication device **14** are wire-connected, wired communication and the wireless communication can be configured to be switched. Even in a case where the first communication device **12** and the second communication device **14** are wire-connected, the operating device **20** and the controller **52** are configured to control the first communication device **12** to send a wireless signal to the second communication device **14**. In this case, electric power can be configured to be supplied from one of the first battery BT**1** and the second battery BT**2** that is electrically connected to the operating device **20** and the first communication device **12**. Wired communication and wireless communication are configured to be switched, for example, by operation of the operating device **20**.

(89) In the communication system **10** of the third embodiment and the communication system **10** of the fourth embodiment, the second communication device **14** and the fourth communication device **18** can be configured to perform wired or wireless communication with each other. In this case, a signal including information that is received by one of the second communication device **14** and the fourth communication device **18** is sent to the other one of the second communication device **14** and the fourth communication device **18**.

(90) FIG. **10** shows a modified embodiment of a communication system **10** in which one of the first battery BT**1** and the second battery BT**2** is configured to supply electric power to the operating device **20**. The first battery BT**1** and the operating device **20** can be configured not to be directly connected by a power line. The first communication device **12** and the second communication device **14** are connected by a power line that allows for power line communication. One of the first battery BT**1** and the second battery BT**2** supplies electric power to the operating device **20** via the transmission device **30**. The second battery BT**2** can be omitted. The first battery BT**1** can be configured to supply electric power to only the component **40**, and the operating device **20** can be configured to be supplied with electric power from only the second battery BT**2** via the transmission device **30**.

(91) In the communication system **10** including the fourth communication device **18**, both the second communication device **14** and the fourth communication device **18** can receive the first information from the first communication device **12**. Alternatively, one of the second communication device **14** and the fourth communication device **18** that performs wired communication with the first communication device **12** can receive the first information from the first communication device **12**. It is preferred that one of the second communication device **14** and the fourth communication device **18** that performs wired communication with the third communication device **16** sends an output to the third communication device **16**.

(92) Even in a case where the communication system **10** is wire-connected, it can be configured to

output a wireless signal. The controller **52** of the first embodiment skips step S**12** and executes step S**14** subsequent to step S**11**. The controller **52** skips step S**15** and executes step S**17**. In the communication system **10** of the second embodiment, the first controller **52**A and the second controller **52**B execute the process in the same manner.

(93) In the second embodiment, one of the first controller **52**A, the second controller **52**B, and the third controller **52**C can be omitted. In a case where the first controller **52**A is omitted, the second controller **52**B controls the first communication device **12** and the second communication device **14**. In a case where the second controller **52**B is omitted, the first controller **52**A controls the first communication device **12** and the second communication device **14**. In a case where the third controller **52**C is omitted, the second controller **52**B controls the second communication device **14** and the third communication device **16**.

(94) In the second embodiment, the second controller **52**B can control the transmission device **30** based on third information. In an example, the third information includes information related to output of the motor unit **42**. The second controller **52**B controls the transmission device **30** to decrease the transmission ratio in a case where the output of the motor unit **42** is large, and controls the transmission device **30** to increase the transmission ratio in a case where the output of the motor unit **42** is small.

(95) In the second embodiment, the second controller **52**B can be configured to skip step S**35** and, subsequent to step S**36**, control the transmission device **30** based on the information related to the transmission device **30** included in the third information.

(96) In this specification, the phrase “at least one of” as used in this disclosure means “one or more” of a desired choice. As one example, the phrase “at least one of” as used in this disclosure means “only one choice” or “both of two choices” in a case where the number of choices is two. In another example, in this specification, the phrase “at least one of” as used in this disclosure means “only one single choice” or “any combination of equal to or more than two choices” if the number of its choices is equal to or more than three.

Claims

1. A component configured to be mounted on a human-powered vehicle, the component comprising: a communication device configured to receive predetermined information sent from a transmission of the human-powered vehicle; and an electronic controller provided on the component the electronic controller being configured to control the component based on whether the predetermined information includes the information related to the component, the predetermined information including at least some of first information that the transmission receives from an operating device mounted to the human-powered vehicle, the operating device including a switch.
2. The component according to claim 1, wherein the communication device is configured to receive the predetermined information on a wireless signal.
3. The component according to claim 1, further comprising a motor unit configured to apply a propulsion force to the human-powered vehicle.
4. The component according to claim 1, wherein the operating device is configured to communicate with the transmission device using a wireless signal.
5. The component according to claim 1, wherein the communication device is connected to the transmission by a wire to communicate with the transmission using a wire signal.
6. The component according to claim 1, wherein the first information includes at least one of information related to the transmission and information related to the component.
7. The component according to claim 6, wherein the information related to the component is different from the information related to the transmission.
8. The component according to claim 1, wherein the predetermined information includes at least

one of information related to the transmission and information related to the component.
9. The component according to claim 8, wherein the information related to the component is different from the information related to the transmission.
10. The component according to claim 1, wherein the first information conforms to the predetermined information.
11. The component according to claim 1, wherein the component is configured to receive a signal sent from the operating device.
12. A communication system comprising the component according to claim 1, and further comprising: at least one battery configured to supply electric power to at least one of the component, the operating device and the transmission.
13. The communication system according to claim 12, wherein the battery is configured to supply electric power to the component and the transmission.
14. The communication system according to claim 13, wherein the battery is configured to supply electric power to the component and the operating device.
15. The communication system according to claim 12, wherein the battery includes: a first battery configured to supply electric power to the transmission device and the component; and a third battery configured to supply electric power to the operating device.
16. A component configured to be mounted on a human-powered vehicle, the component comprising: a communication device configured to receive predetermined information sent from a transmission of the human-powered vehicle; and an electronic controller configured to control the component based on the predetermined information in a state where the predetermined information includes information related to the component, the predetermined information including at least one of information related to the transmission and information related to the component that the transmission receives from an operating device mounted to the human-powered vehicle, the information related to the transmission being different from the information related to the component, the operating device including a switch.
17. The component according to claim 16, wherein the information related to the transmission includes at least one of information related to a control state of the transmission and information for controlling the transmission, and the information related to the component includes at least one of information related to a control state of the component and information controlling the component.
18. The component according to claim 17, wherein the component is a motor unit configured to apply a propulsion force to the human-powered vehicle, the information related to the control state of the transmission includes information indicating a current transmission ratio, the information for controlling the transmission includes information of an operating signal related to the changing the transmission ratio, the information related to the control state of the component includes a first control state and a second control state that differs in assisting force from the first control state, and the information related to controlling the component includes information related to assisting force.
19. The component according to claim 16, wherein the electronic controller is configured to determine whether the predetermined information includes the information related to the component and control the component based on the predetermined information only upon determining that the predetermined information includes the information related to the component.